EXPERIMENT 9

Student Name: Aditya Verma **UID: 24BAI70937**
Branch: AIT-CSE (AIML) **Section/Group: 24AIT_KRG-G2**
Semester: 4
Subject Name: Database Management System **Subject Code: 24CSH-298**

Experiment 9 – Creating and Implementing Packages in PL/SQL

Experiment 9: PL/SQL Package Development: Procedures and Shared Cursors for Enterprise

Aim

To create and implement PL/SQL packages by developing a package specification and package body containing procedures and shared cursors, in order to achieve modular, reusable, and efficient database programming.

Objective

- To understand the concept and structure of PL/SQL packages.
- To differentiate between package specification and package body.
- To implement procedures inside packages.
- To use shared cursors for retrieving data.
- To develop modular and reusable PL/SQL code.

Software Requirements

Database Management System:

- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

Database Administration Tool / Client Tool:

- Oracle SQL Developer
- pgAdmin

EXPERIMENT 9

Problem Statement

In enterprise applications, database operations are grouped into packages for better maintainability, security, and reuse. A structured package should be designed to fetch and display employee details using shared cursors.

Practical / Experiment Steps

- Create an employee table.
- Insert sample data.
- Create a package specification.
- Create a package body.
- Define procedures inside the package.
- Use a shared cursor to fetch employee records.
- Execute package procedures.
- Display output.

Procedure

- ☐ Open Oracle SQL Developer.
- ☐ Create employee table.
- ☐ Insert sample data.
- ☐ Create package specification.
- ☐ Create package body.
- ☐ Declare shared cursor.
- ☐ Write procedure to fetch employee details.
- ☐ Compile package.
- ☐ Execute procedure.
- ☐ Display output.

EXPERIMENT 9

Input / Output Details

Input

- Table: employee (id, name, salary)

Step-wise Output

Step 1 – Create Employee Table

```
CREATE TABLE employee (
    id NUMBER PRIMARY KEY,
    name VARCHAR2(50),
    salary NUMBER
);
```

Step 2 – Insert Sample Data

```
-- Step 2: Insert Sample Data
INSERT INTO employee VALUES (1, 'Rahul', 50000);
INSERT INTO employee VALUES (2, 'Amit', 60000);
INSERT INTO employee VALUES (3, 'Neha', 55000);

COMMIT;
```

Step 3 – Create Package Specification

```
-- Step 3: Package Specification
CREATE OR REPLACE PACKAGE emp_package AS
    PROCEDURE display_employees;
END emp_package;
/
```

Step 4 – Create Package Body

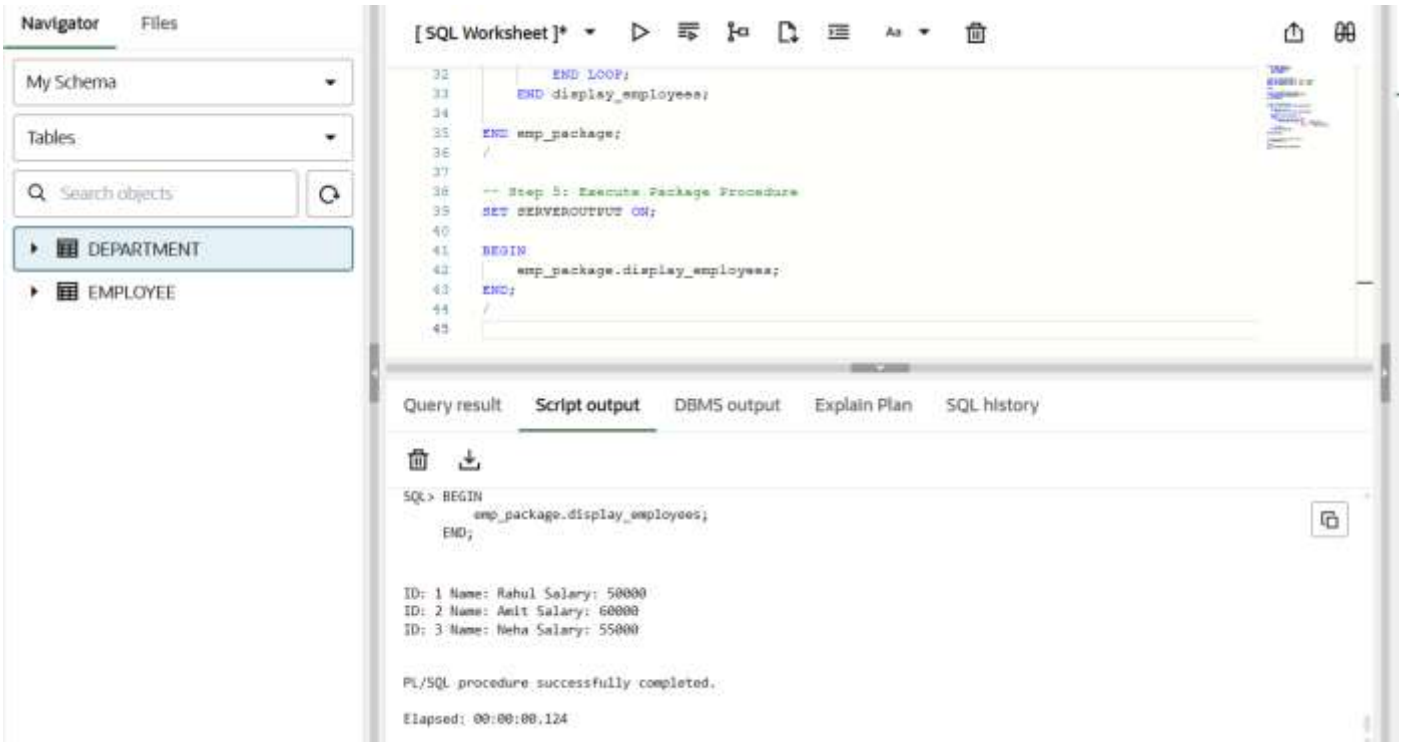
EXPERIMENT 9

```oracle_plsql
-- Step 4: Package Body
CREATE OR REPLACE PACKAGE BODY emp_package AS

    CURSOR emp_cursor IS
        SELECT id, name, salary FROM employee;

    PROCEDURE display_employees IS
    BEGIN
        FOR emp_rec IN emp_cursor LOOP
            DBMS_OUTPUT.PUT_LINE('ID: ' || emp_rec.id ||
                                 ' Name: ' || emp_rec.name ||
                                 ' Salary: ' || emp_rec.salary);
        END LOOP;
    END display_employees;
```

Step 5 – **Execute Package Procedure**



EXPERIMENT 9

Learning Outcome

After completing this experiment, the learner will be able to:

- Understand PL/SQL package structure.
- Implement package specification and body.
- Use shared cursors effectively.
- Develop modular and reusable code.
- Apply enterprise-level database programming practices used in companies like Rippling, Oracle, SAP, and PayPal.